WIPE TEST

Leak Test: an essential control for sealed radioactive sources



Leak Test: an essential control for sealed radioactive sources

Sealed radioactive sources are used in different sectors of industry, medicine, research and specialized services, in equipment, processes and technical applications. Although they are designed to operate safely, their integrity must be verified periodically.

This is where the Leak Test becomes an important step in radiation protection routines. The test aims to assess whether there are signs of radioactive material being released from a sealed source, helping to verify that the source remains intact, with no signs of leakage or contamination on accessible surfaces.

How to choose the best way to perform the Leak Test

Although it may seem like a simple procedure, the Leak Test involves technical requirements that must be met to ensure reliable results, such as:

- choice of location and sampling method;
- definition of the type of test, such as dry Wipe Test, wet Wipe Test or liquid scintillation immersion;
- detection limits to be met, according to ISO 9978:2020;
- acceptance criteria to consider that the source is free of leakage.

It is therefore essential that the test be performed by a qualified technical team, using appropriate methodologies and in compliance with the requirements of ISO/IEC 17025:2017, ensuring traceability, reliability and safe support for decision-making by the Radiation Protection Service.



Why is this control necessary?

Sealed radioactive sources are designed to keep the radioactive material confined. However, factors such as time in use, operating conditions, transport, storage, physical wear or encapsulation failures may call for additional controls to verify the integrity of the source.

The Leak Test acts as a preventive measure. It makes it possible to identify possible signs of contamination before the situation poses greater risks to workers, the public, the environment or operational continuity.

More than a technical requirement, this control is directly linked to radiological safety, contamination prevention and compliance with the requirements applicable to facilities that use radioactive sources.

Where can this test be applied?

The Leak Test may be necessary in different contexts, such as:

- industries that use nuclear level, density or thickness gauges;
- fixed process equipment and industrial irradiators;
- laboratories, metrology centers and calibration sources;
- research institutions;
- medical services that use sealed sources in specific applications.

The frequency and criteria must be aligned with the facility's Radiation Protection Plan and with the requirements applicable to each type of source and operation.

How is the immersion test performed?

The liquid scintillation immersion test is a technique used mainly to assess removable surface contamination on small parts, filters, tools, PPE, environmental samples or surfaces with complex geometry that may be contaminated with low-energy beta emitters.

The principle consists of transferring the radionuclides present on the surface to a scintillation solution and measuring the activity by Liquid Scintillation Counting, known as LSC.

When the part cannot be immersed directly in the scintillation liquid, a common practice is to perform the Wipe Test and then immerse in the scintillation liquid the filter or swab used for sampling.

How is the Wipe Test performed?

In general, the Wipe Test consists of collecting samples by controlled rubbing with absorbent material at specific points on the source or equipment.

Sampling must consider factors such as accessibility, shielding, geometry and operating conditions. The collected material is then sent for radiometric analysis in a specialized laboratory.

When performed correctly, the Wipe Test helps reduce risks, identify possible nonconformities and strengthen the operational control of the facility.

Safety, traceability and reliability

Because it involves the assessment of radioactive material, the Leak Test requires technical care at every stage: sampling planning, execution of the procedure, sample transport, laboratory analysis and interpretation of results.

The reliability of the test depends on the technical competence involved and on the use of appropriate methodologies. Therefore, relying on specialized, accredited laboratories is an important factor in ensuring traceability, analytical quality and safe decision-making.

How GLP Laboratórios can help

GLP Laboratórios works with radiometric analyses focused on monitoring radioactive contamination and controlling sources, providing technical support to companies, industries, laboratories and institutions that need to operate safely and reliably.

With a specialized structure, a qualified technical team and methodologies suited to the sector's demands, GLP helps ensure your operation has reliable, traceable results aligned with applicable requirements.

If your company uses sealed radioactive sources, the Leak Test may be an essential step to strengthen the radiological control of your operation.

Contact us

WhatsApp
+55 19 98455-9001

Phone
+55 19 3413-5437

E-mail
propostas@glplab.com.br

vendas@glplab.com.br